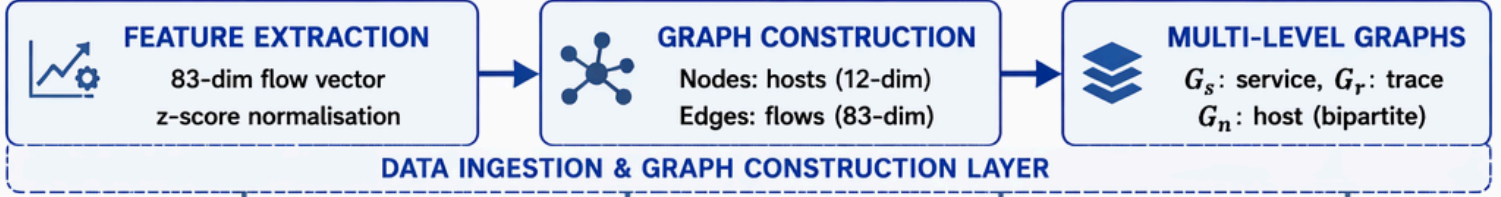




RAW NETWORK TRAFFIC

PCAP / NetFlow / CSV / Syslog / API feeds





CONDITION 1

Adversarial Environment



CONDITION 2

Distributed Dynamics



CONDITION 3

Non-Stationarity



R1: ROBUSTNESS



R2: ACCURACY



R3: EFFICIENCY



R4: PRIVACY

- M1: CT-TGNN
- M4: MambaShield
- M7: FedGTD Cert.

- M5: Stoch. Transformer

- M4: MambaShield

- M7: FedGTD Cert.

- M1: CT-TGNN

- M3: TripleE-TGNN

- M4: MambaShield

- M2: FedLLM-API

- M6: UC-HGP

- M5: Stoch. Transformer

- M4: MambaShield

- M7: FedGTD Cert.



FEDERATED LEARNING LOOP

 SOC 1 SOC 2 SOC 3
...

▽ Gradient upload
Model download ↓

DP noise
OT alignment



CONTINUAL LEARNING LOOP

EWC Fisher
Replay buffer
Drift adaptation



DECISION ENGINE

Game-theoretic strategy → Uncertainty triage ($\tau = 0.3$) → Severity ranking → MITRE ATT&CK mapping

CONFIDENCE-SCORED ALERTS

Critical / High / Medium / Low / Informational | with epistemic uncertainty U_{epi} , U_{ale}



RobustIDPS.ai PLATFORM

46 pages | 9 groups | 13+ models | SOC Intelligence | LLM Defence (94.5%) | Snort/Suricata plug-in

LEGEND

- Certified dynamics
- Efficiency / uncertainty
- Certification
- Foundation model
- Post-quantum
- > Dependencies

3 CONDITIONS (Adversarial, Distributed, Non-Stationary) × 4 REQUIREMENTS (Robustness, Accuracy, Efficiency, Privacy) = 12 / 12 CELLS VERIFIED